

1. Express, $\sqrt[3]{81^3} \times \sqrt{27}$ with base 27.

$$\begin{aligned}
 &= (3^4)^{\frac{3}{5}} \times (3^3)^{\frac{1}{2}} \\
 &= 3^{\frac{12}{5}} \times 3^{\frac{3}{2}} \\
 &= 3^{\frac{39}{10}}
 \end{aligned}
 \quad \Rightarrow \quad = 27^{\frac{13}{10}}$$

2. Express as a single logarithm in the form $\log_b x$, and state restrictions if necessary.

a) $\log_3 12 + \frac{1}{2} \log_3 4 - \log_3 6$

$$= \log_3 12 + \log_3 4^{\frac{1}{2}} - \log_3 6$$

$$= \log_3 \left(\frac{12 \times 2}{6} \right)$$

$$= \log_3 4$$

b) $3 \log_4 x + 2 \log_4 x - \log_4 y$

$$= \log_4 x^3 + \log_4 x^2 - \log_4 y$$

$$= \log_4 \left(\frac{x^3 x^2}{y} \right)$$

$$= \log_4 \left(\frac{x^5}{y} \right)$$

$$x > 0, y > 0$$

c) $\log(x^2 - x - 6) - \log(3x - 9)$

$$= \log[(x-3)(x+2)] - \log[3(x-3)]$$

$$= \log(x-3) + \log(x+2) - \log 3 - \log(x-3)$$

$$= \log \left(\frac{x+2}{3} \right)$$

$$x > 3$$

d) $\log_{32} \sqrt{x^2 - 3x} = \frac{1}{5}$

$$\frac{1}{2} \log_{32} (x^2 - 3x) = \frac{1}{5}$$

$$\log_{32} (x^2 - 3x) = \frac{2}{5}$$

$$x^2 - 3x = 32^{\frac{2}{5}}$$

$$x^2 - 3x = 4$$

$$x^2 - 3x - 4 = 0$$

$$(x-4)(x+1) = 0$$

$$x = -1, 4$$

3. Solve the following equations using logarithm laws.

a) $6^{3x+1} = 2^{2x-3}$

$$(3x+1)\log 6 = (2x-3)\log 2$$

(6) $3x\log 6 + \log 6 = 2x\log 2 - 3\log 2$

$$3x\log 6 - 2x\log 2 = -\log 6 - 3\log 2$$

$$x(3\log 6 - 2\log 2) = -\log 6 - 3\log 2$$

$$x = \frac{-\log 6 - 3\log 2}{3\log 6 - 2\log 2}$$

$$= \frac{-\log(6 \cdot 8)}{\log \frac{216}{4}}$$

Exact: $\log_{54} \left(\frac{1}{48} \right)$

Approximate (2 decimal places): -0.97

b) $2(7)^{4x-5} = 30$

$$7^{4x-5} = 15$$

$$(4x-5)\log 7 = \log 15$$

$$4x-5 = \log_7 15$$

$$x = \frac{1}{4}(\log_7 15 + 5)$$

$$= \frac{1}{4}(\log_7 15 + \log_7 7^5)$$

$$= \log_7 (15 \times 7^5)^{\frac{1}{4}}$$

$$x = \log_7 \sqrt[4]{252105}$$

4. Solve the following equations and state any extraneous roots if they exist. State restrictions, when and if possible. Round to one decimal place.

a) $2^x - 4 + 3(2^{-x}) = 0$

$$(2^x)^2 - 4(2^x) + 3 = 0$$

(6) $(2^x - 3)(2^x - 1) = 0$

$$2^x = 3 \quad 2^x = 1$$

$$x = \log_2 3 \quad x = 0$$

$$\approx 1.6$$

b) $\log_3(x+4) + \log_3(2x-3) = \log_3 6$

$$\log_3[(x+4)(2x-3)] = \log_3 6$$

$$2x^2 + 5x - 12 = 6$$

$$2x^2 + 5x - 18 = 0$$

$$(2x+9)(x-2) = 0$$

$$x = -\frac{9}{2} \text{ or } 2$$

DNE

since log of

negative

number

5. The mass of a radioactive substance is 500g from the beginning and only 60g remains after 40 hours. Determine, to four decimal places:

a) the half life.

b) the amount that remains after 3 days.

$$60 = 500 \left(\frac{1}{2} \right)^{\frac{40}{h}}$$

$$\frac{3}{25} = \left(\frac{1}{2} \right)^{\frac{40}{h}}$$

$$\frac{40}{h} = \log_{\frac{1}{2}} \left(\frac{3}{25} \right)$$

$$h = \frac{40}{\log_{\frac{1}{2}} \left(\frac{3}{25} \right)} \approx 13.0766 \text{ hours}$$

$$3 \text{ days} \times \frac{24 \text{ hours}}{1 \text{ day}} = 72 \text{ hours}$$

$$A(72) = 500 \left(\frac{1}{2} \right)^{\frac{72}{13.0766}}$$

$$\approx 11.0025 \text{ g}$$

6. Mr. Azar wants to buy a new car that costs \$80,000. He invested \$61,000 at 6.2%/annum compounded semi-annually. How long will he have to wait to buy this new car?

$$A = P \left(1 + \frac{i}{n} \right)^{nt}$$

$$80000 = 61000 \left(1 + \frac{0.062}{2} \right)^{2t}$$

$$\frac{80}{61} = (1.031)^{2t}$$

$$t = \frac{1}{2} \log_{1.031} \left(\frac{80}{61} \right)$$

$$t = 4.44 \text{ years}$$

$$\Rightarrow 4 \text{ years and 6 months}$$

7. Solve, $\log_3 x + \log_{27} x + \log_{729} x = 14$, for x . Round to the nearest whole number.

$$\frac{\log x}{\log 3} + \frac{\log x}{\log 27} + \frac{\log x}{\log 729} = 14$$

$$\frac{\log x}{\log 3} + \frac{\log x}{3 \log 3} + \frac{\log x}{6 \log 3} = 14$$

$$\frac{6 \log x}{6 \log 3} + \frac{2 \log x}{6 \log 3} + \frac{\log x}{6 \log 3} = 14$$

$$\frac{\log x^6 + \log x^2 + \log x}{\log 3^6} = 14$$

$$\frac{\log x^9}{\log 3^6} = \log_{3^6} (3^6)^{14}$$

$$x^9 = 3^{84}$$

$$x = 3^{\frac{84}{9}}$$

$$x = 3^{\frac{28}{3}}$$

$$x \approx 28388$$

8. If $\left(\frac{1}{2}\right)^{x+y} = 16$ and $\log_{x-y} 8 = -3$, calculate the values of x and y . Leave answers in exact form.

$$\left(\frac{1}{2}\right)^{x+y} = 16$$

$$\log_{x-y} 8 = -3$$

$$\textcircled{1} + \textcircled{2} : 2x = -\frac{7}{2}$$

$$x+y = \log_{\frac{1}{2}}(16)$$

$$(x-y)^3 = 8$$

$$x = -\frac{7}{4}$$

$$x+y = \log_{\frac{1}{2}}\left(\frac{1}{2}\right)^{-4}$$

$$(x-y)^3 = \frac{1}{8}$$

$$y = -4 - \left(-\frac{7}{4}\right)$$

$$\textcircled{1} x+y = -4$$

$$\textcircled{2} x-y = \frac{1}{2}$$

$$= -\frac{9}{4}$$

9. Given, $2(5^{4x}) - 9(5^{2x}) + 10(5^{2x}) - 3 = 0$, solve for x . Round to three decimal places.

$$\begin{array}{r|rrrr} 1 & 2 & -9 & 10 & -3 \\ & \downarrow & 2 & -7 & 3 \\ \hline & 2 & -7 & 3 & 0R \end{array}$$

$$2(5^{2x}) - 1 = 0$$

$$5^{2x} - 3 = 0$$

$$5^{2x} = \frac{1}{2}$$

$$5^{2x} = 3$$

$$x = \frac{1}{2} \log_5 \left(\frac{1}{2}\right)$$

$$x = \frac{1}{2} \log_5 3$$

$$= -0.215$$

$$= 0.341$$

$$= (2(5^{4x}) - 7(5^{2x}) + 3)(5^{2x} - 1)$$

$$= [2(5^{2x}) - 1][5^{2x} - 3](5^{2x} - 1)$$

$$\begin{array}{l} 5^{2x} - 1 = 0 \\ 5^{2x} = 1 \end{array} \quad \begin{array}{l} 2x = 0 \\ x = 0 \end{array}$$

10. Given $2\log_4 x + 2\log_x 4 = 5$, solve for x . Must show work for full marks.

$$\frac{2\log x}{\log 4} + \frac{2\log 4}{\log x} = 5$$

$$\log x = \frac{5\log 4 \pm \sqrt{(-5\log 4)^2 - 4(2)(2(\log 4)^2)}}{2(2)}$$

$$2(\log x)^2 + 2(\log 4)^2 = 5\log 4 \log x$$

$$= 0.301 \text{ or } 1.204$$

$$2(\log x)^2 - 5\log 4 \log x + 2(\log 4)^2 = 0$$

$$x = 10^{0.301} \text{ or } 10^{1.204}$$

$$x = 2 \text{ or } 16$$